

Teorema de aproximación de la identidad por convolución

Teorema 1. Sea $\phi \in \mathcal{L}^1(\mathbb{R}^n)$ tal que $\phi \geq 0$ y $\int_{\mathbb{R}^n} \phi(x) dx = 1$. Entonces,

- (1) $\forall f \in \mathcal{L}^p(\mathbb{R}^n)$ con $p \in [1, \infty)$: $\lim_{t \rightarrow 0^+} \|f * \phi_t - f\|_p = 0$.
- (2) $\forall f \in \mathcal{L}^\infty(\mathbb{R}^n)$ uniformemente continua : $\lim_{t \rightarrow 0^+} \|f * \phi_t - f\|_\infty = 0$.
- (3) $\forall f$ continua en $x_0 \in \mathbb{R}^n$: $\lim_{t \rightarrow 0^+} f * \phi_t(x_0) = f(x_0)$.

Demostración: Como $\int_{\mathbb{R}^n} \phi(x) dx = 1$, por la obs-propiedades-dilatacion-isotropica/??, sabemos que $\forall t > 0$: $\int_{\mathbb{R}^n} \phi_t(x) dx = 1$. Entonces, $\forall x \in \mathbb{R}^n$:

$$\begin{aligned} |f * \phi_t(x) - f(x)| &= \left| \int_{\mathbb{R}^n} f(x-y) \phi_t(y) dy - \int_{\mathbb{R}^n} f(x) \phi_t(y) dy \right| \\ &= \left| \int_{\mathbb{R}^n} [f(x-y) - f(x)] \phi_t(y) dy \right| \leq \int_{\mathbb{R}^n} |T_y f(x) - f(x)| \phi_t(y) dy. \end{aligned} \quad (1)$$

(1) Sea q el exponente conjugado de p (i.e. $1/p + 1/q = 1$). Entonces, por Hölder,

$$\begin{aligned} |f * \phi_t(x) - f(x)| &\stackrel{(\text{??})}{\leq} \int_{\mathbb{R}^n} |T_y f(x) - f(x)| \phi_t^{1/p}(y) \phi_t^{1/q}(y) dy \\ &\stackrel{\text{Hölder}}{\leq} \left(\int_{\mathbb{R}^n} |T_y f(x) - f(x)|^p \phi_t(y) dy \right)^{1/p} \cdot \underbrace{\left(\int_{\mathbb{R}^n} \phi_t(y) dy \right)^{1/q}}_1. \end{aligned} \quad (2)$$

$$\begin{aligned} \|f * \phi_t - f\|_p^p &= \int_{\mathbb{R}^n} |f * \phi_t(x) - f(x)|^p dx \stackrel{(\text{??})}{\leq} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |T_y f(x) - f(x)|^p \phi_t(y) dy dx = \\ &\stackrel{\text{Fubini}}{\leq} \int_{\mathbb{R}^n} \phi_t(y) \left(\int_{\mathbb{R}^n} |T_y f(x) - f(x)|^p dx \right) dy = \int_{\mathbb{R}^n} \phi_t(y) \|T_y f - f\|_p^p dy. \end{aligned} \quad (3)$$

Por el lem-convergencia-lp-traslacion/??, dado $\varepsilon > 0$, existe $\eta > 0$ tal que $\forall y \in \mathbb{R}^n$: $\|y\| < \eta$: $\|T_y f - f\|_p < \varepsilon$. Entonces, sea $\eta > 0$ tal que $\|y\| < \eta \implies \|T_y f - f\|_p \leq \frac{\varepsilon}{2^{1/p}}$. Por la obs-propiedades-dilatacion-isotropica/??, $\exists \delta > 0$: $\forall t \in (0, \delta)$:

$\int_{\|y\| \geq \eta} \phi_t(y) \, dy < \frac{\varepsilon^p}{2^{p+1} \|f\|_p^p}$. Entonces, para $t \in (0, \delta)$,

$$\begin{aligned} \|f * \phi_t - f\|_p^p &\stackrel{(\text{??})}{\leq} \int_{\|y\| < \eta} \|T_y f - f\|_p^p \phi_t(y) \, dy + \int_{\|y\| \geq \eta} \|T_y f - f\|_p^p \phi_t(y) \, dy \\ &\leq \frac{\varepsilon^p}{2} \int_{\|y\| \leq \eta} \phi_t(y) \, dy + 2^p \|f\|_p^p \frac{\varepsilon^p}{2^{p+1} \|f\|_p^p} \leq \frac{\varepsilon^p}{2} + \frac{\varepsilon^p}{2} = \varepsilon^p. \end{aligned}$$

Por tanto, $\lim_{t \rightarrow 0^+} \|f * \phi_t - f\|_p = 0$.

(2) Como f es uniformemente continua, $\forall \varepsilon > 0 : \exists \eta > 0 : \forall x, y \in \mathbb{R}^n : \|y\| < \eta \implies |f(x - y) - f(x)| < \frac{\varepsilon}{2}$. Entonces, por (??), tenemos que

$$\begin{aligned} |f * \phi_t(x) - f(x)| &\stackrel{(\text{??})}{\leq} \int_{\|y\| < \eta} |T_y f(x) - f(x)| \phi_t(y) \, dy + \int_{\|y\| \geq \eta} |T_y f(x) - f(x)| \phi_t(y) \, dy \\ &\leq \frac{\varepsilon}{2} \int_{\mathbb{R}^n} \phi_t(y) \, dy + 2 \|f\|_\infty \int_{\|y\| \geq \eta} \phi_t(y) \, dy \leq \frac{\varepsilon}{2} + 2 \|f\|_\infty \frac{\varepsilon}{4 \|f\|_\infty} = \varepsilon. \end{aligned}$$

Por tanto, $\lim_{t \rightarrow 0^+} \|f * \phi_t - f\|_\infty = 0$.

(3) Como f es continua en x_0 , tenemos que

$$\forall \varepsilon > 0 : \exists \eta > 0 : \forall y \in \mathbb{R}^n : \|y\| < \eta \implies \int_{\|y\| < \eta} \phi_t(y) \, dy \leq \frac{\varepsilon}{4 \|f\|_\infty}.$$

Entonces, el resultado se sigue igual que en (2) reemplazando x por x_0 . ■

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